

# Python for Econometrics and Operations Research

A crash course

08-09-2026



# Desktop computer

- ① Start computer by pressing bottom-right button (below Intel vPRO sticker)
- ② Log in with your student number (e.g., u123456) and your password.

# Team



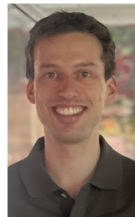
Dr.ir. Sander Gribbling



Dr.ir. Pieter Kleer



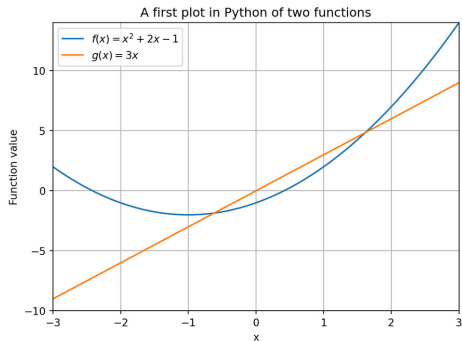
Prof.dr. Johan van Leeuwen



Dr.mr. Sven Polak

# What is Python?

- Python is a (*computational*) *programming language*.
- We instruct a computer to perform operations via written text.
- The text needs to be very exact.
  - ▶ Otherwise the computer will throw a syntax (i.e. spelling/grammar) error.



# About Python

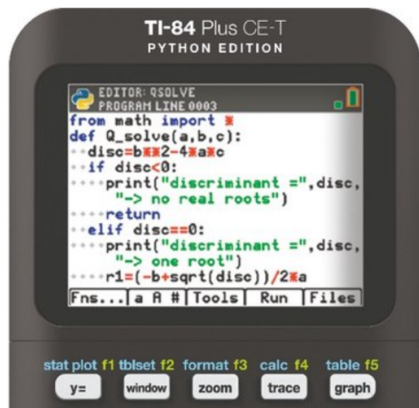
Popular programming language for **data science tasks**.

- Analysis of mathematical **high-dimensional functions**.
  - ▶ Plotting, finding roots/minima/maxima, integration, etc.
- Analysis and visualization of **large-scale data sets**.
  - ▶ E.g., weather or stock market predictions.



| MON        | TUE        | WED        | THU        | FRI        | SAT        | SUN        |
|------------|------------|------------|------------|------------|------------|------------|
|            |            |            |            |            |            |            |
| 29°<br>15° | 22°<br>15° | 16°<br>14° | 19°<br>13° | 21°<br>12° | 18°<br>12° | 20°<br>11° |
| 30 Jun     | 1 Jul      | 2 Jul      | 3 Jul      | 4 Jul      | 5 Jul      | 6 Jul      |

# Graphing calculator vs. Python



Graphing calculator can analyse one-dimensional functions like

$$f(x) = x^2 + 2x - 1.$$

Python can analyse higher-dimensional functions like

$$f(x, y) = x^2y + 2xy^2 - x - 1.$$

# Why this programming language?

Free, open-source and most popular programming language for data science!

Many companies program in Python ...

- ... including many master thesis students who do company internships.

Good skill to have on you CV!



## Other languages in curriculum

You will also see other programming languages like Matlab and R:



- Similar functionality as Python (although “syntax” is different).

AI-tools like ChatGPT can also program! Not always allowed, though.

# Python (and programming) in EOR curriculum

Programming courses:

- Programming for EOR (Year 2)
- Computational Aspects in Econometrics (elective, Year 3)

Assignments of:

- Linear Optimization (**Q2**), Probability Theory, Introduction Finance and Actuarial Sciences, Quantative Finance, ....

# Plan for today

Materials (online book) at: <https://pskleer.github.io/eor-python-crash-course-2025/>

- Cover **Python Basics (Chapter 3)** using centralized explanations and **Exercises 3.1-3.7**.
- Lecture 2 (September 22) will cover Linear Algebra with Python!

## Chapter 3 of online "book" - Python basics

## Python as a calculator

Can use +, -, \*, /, and \*\* to perform basic arithmetic operations.

| Operation      | Symbol | Example  | Result |
|----------------|--------|----------|--------|
| Addition       | +      | $2 + 3$  | 5      |
| Subtraction    | -      | $5 - 3$  | 2      |
| Multiplication | *      | $2 * 3$  | 6      |
| Division       | /      | $3 / 2$  | 1.5    |
| Exponentiation | **     | $2 ** 3$ | 8      |

# Variables

Suppose we want to compute the function value  $f(x) = x^3 + 2x^2 + x - 1$  for  $x = 5$ .

```
5**3 + 2*5**2 + 5 - 1
```

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If we want  $x = 4$ , we have to change 5 to 4 in three places. Better to define variable for  $x$ .

```
x = 5
```

```
x**3 + 2*x**2 + x - 1
```

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**Try Exercises 3.1 and 3.2!**

# Lists

Can store multiple variables in a *list* object.

```
z = [3, 9, 1, 7]
```

```
z
```

```
[3, 9, 1, 7]
```

Can access numbers in list by *indexing* them.

```
z[1]
```

```
9
```

Why does this give 9? **Python starts counting at 0 when indexing.**

```
z[0]
```

```
3
```

## For-loop

```
a = [1, 4, 2, 5]

total_sum = a[0] + a[1] + a[2] + a[3]
total_sum
```

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More efficient to use a *for-loop* for adding up numbers in *a*.



## For-loop (cont'd)

```
a = [1, 4, 2, 5]
```

```
total_sum = 0
```

```
for i in [0,1,2,3]:
```

```
    total_sum = total_sum + a[i]
```

```
print(total_sum)
```

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| i | a[i] | total_sum after this iteration |
|---|------|--------------------------------|
| 0 | 1    | $0 + 1 = 1$                    |
| 1 | 4    | $1 + 4 = 5$                    |
| 2 | 2    | $5 + 2 = 7$                    |
| 3 | 5    | $7 + 5 = 12$                   |

## For-loop (cont'd)

Suppose we only want the first two numbers in  $a$ : Change list with values for  $i$ .

```
a = [1, 4, 2, 5]
total_sum = 0

for i in [0,1]:
    total_sum = total_sum + a[i]

print(total_sum)
```

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**Try Exercises 3.3 and 3.4!**

# Conditional statements

Carry out different lines of code depending on what statement/condition is true.

- Can be done with `if` and `else` keywords.

```
x = 5

if x > 0:
    print("x is positive")
else:
    print("x is is not positive")
```

x is positive

## Conditional statements (cont'd)

If we have more than three conditions, we can use `if`, `elif` and `else`

```
x = 0

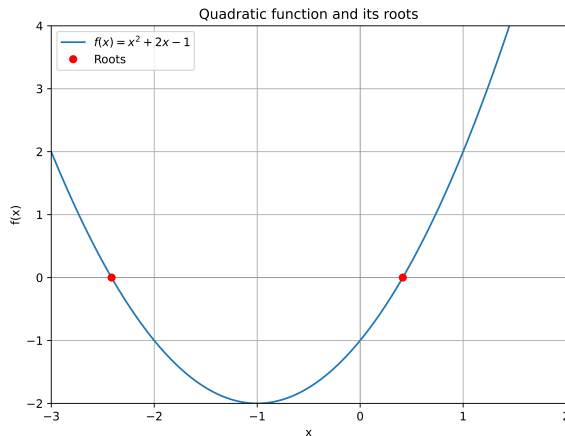
if x > 0:
    print("x is positive")
elif x == 0:
    print("x is zero")
else:
    print("x is negative")
```

x is zero

**Try Exercise 3.5!**

## Mathematical example: Root finding for quadratic function.

Let  $a, b$  and  $c$  be given. Find  $x$  such that  $f(x) = a \cdot x^2 + b \cdot x + c = 0$ .



# Discriminant

Number of solutions to  $f(x) = a \cdot x^2 + b \cdot x + c = 0$  determined by **discriminant**  $D = b^2 - 4ac$ :

- If  $D > 0$ , the equation has two real roots  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- If  $D = 0$ , the equation has exactly one real root  $x = \frac{-b}{2a}$ .
- If  $D < 0$ , there are no real roots.

**Try Exercise 3.6 (and 3.7)!**